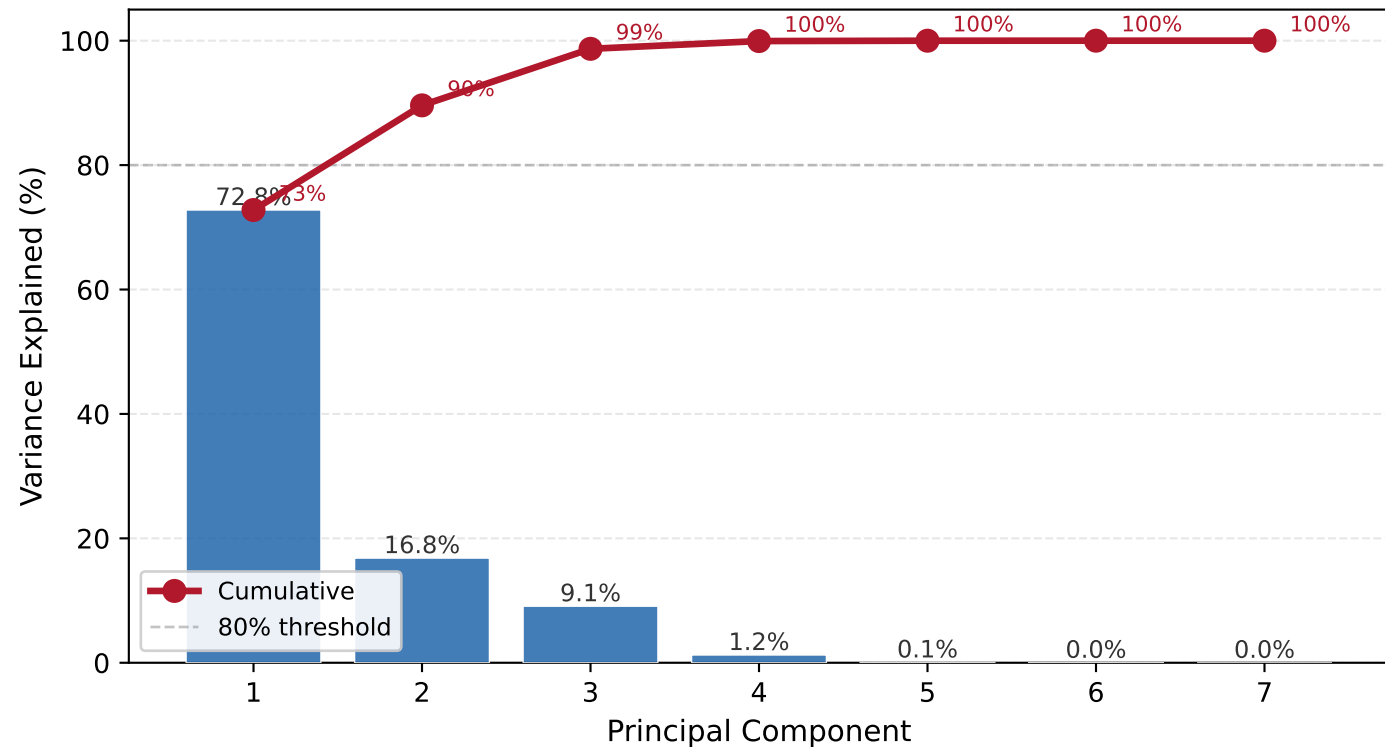


PCA Scree Plot: How Many Dimensions Matter?



IMPLICATIONS: PC1 (72.8%) = physical scale & performance. PC2 (16.9%) = rotor count. PC3 (9.1%) = elevation angle. First 2 PCs capture 89.7%; 3 PCs capture 98.8%. The 7-D BEM design space is notably compressible — most engineering PCA needs 4-5 components to hit 90%. Funding implication: optimize radius (PC1), choose stack count (PC2), fine-tune elevation (PC3). Tilt profile contributes noise. Data: 96 configurations x 7 variables from BEM v2.0 sweep.